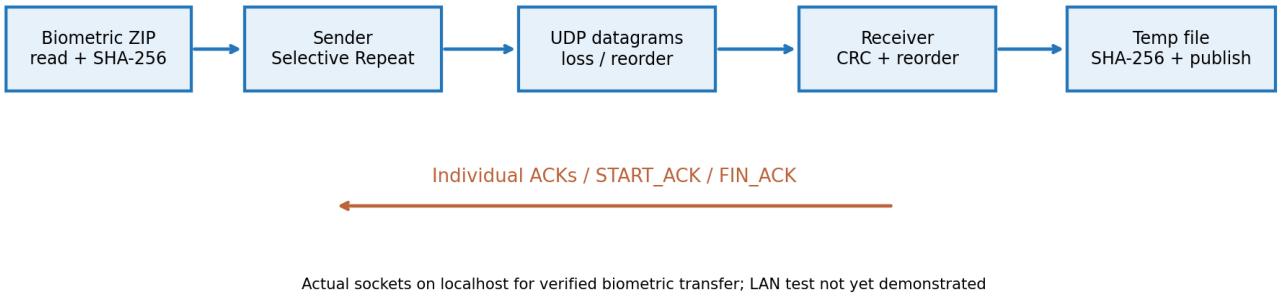
BioNet: Reliable Biometric Archive Transfer Using Selective Repeat ARQ over UDP

Objective. Implement a genuine networking mechanism that transfers captured biometric data, reconstructs the exact bytes, verifies integrity, and evaluates performance. The selected mechanism is a custom Selective Repeat Automatic Repeat reQuest (ARQ) protocol over UDP; instructor approval remains to be confirmed.

Architecture and protocol



Actual sockets on localhost for verified biometric transfer; LAN test not yet demonstrated

Python 3.12 uses real UDP sockets and a BioNet v1 frame with a 32-byte header (magic, version, type, session UUID, 64-bit sequence, payload length) and 4-byte CRC32. Default DATA chunks contain 1,024 bytes. START carries filename, length, expected SHA-256, chunk count and proposed window; START_ACK negotiates a bounded window. FIN_ACK follows successful receiver-side validation.

The sender maintains individual packet timers, per-packet ACKs and selective retransmissions. If packets 0, 2 and 3 are ACKed but 1 is missing, the base remains at 1 and only unacknowledged packet 1 is retransmitted. The receiver buffers out-of-order packets, writes contiguous chunks to a temporary file, updates SHA-256, and publishes only verified output without overwriting existing files. CRC32 detects accidental datagram corruption; SHA-256 checks complete-file equality. Neither authenticates or encrypts biometric data.

Implementation and evaluation method

Modules include **protocol, sender, receiver, impairment, integrity, dashboard, TCP baseline and benchmark**. The optional Rich dashboard displays window position, ACKs, retransmissions, progress and verification, without showing biometric contents. Tests use synthetic data; benchmark timing excludes source pre-hashing and receiver startup. Goodput = (file bytes × 8) / (completion seconds × 10■). Synthetic benchmark conditions: Linux localhost, 65,536-byte payload, window 32, three trials per loss setting, independent loss at both endpoints and 2 ms sender delay.

Measured results and verification

Table 1. **Phase 3 synthetic UDP sweep** (65,536 bytes, window 32, 3 trials per setting). Values are recorded measurements, not estimates.

Loss / direction	Mean goodput	Mean time	Retries (3 trials)
0%	22.596 Mbps	0.023 s	0
5%	1.830 Mbps	0.295 s	25
10%	1.299 Mbps	0.465 s	56
20%	0.660 Mbps	0.805 s	124
30%	0.436 Mbps	1.247 s	210
40%	0.252 Mbps	2.099 s	335

A separate no-impairment 64 KiB localhost comparison measured mean goodput of **743.686 Mbps for TCP**, **174.459 Mbps for UDP window 8** and **182.180 Mbps for UDP window 32** (three trials each). These are implementation-specific localhost results, not evidence of general transport superiority.

Table 2. **Actual biometric archive transfer on MacBook Air M4**, localhost UDP, sender configured with 5% simulated outbound loss.

Metric	Observed result
Archive size	2,774,235,808 bytes (2.77 GB)
Completion time	6,453.900 seconds (1 h 47 min 34 s)
DATA packets ACKed	2,709,215 / 2,709,215
Retransmissions / simulated sender drops	143,000 / 143,000
Receiver out-of-order arrivals	1,857,096
Final sender-reported goodput	3.439 Mbps
Integrity	Sender SHA-256 PASS; receiver SHA-256 PASS; independent cmp PASS

Validation and conclusion

On the actual Mac (Python 3.12.8), **181/181 pytest tests passed in 30.79 s**, and the macOS synthetic validation script passed. A 1.05 MB synthetic ZIP transfer also passed SHA-256 with 58 retries. The real 2.77 GB biometric ZIP was reconstructed successfully despite simulated loss; an independent byte-for-byte `cmp` comparison confirmed identical output.

Limitations: real biometric transfer was **localhost, not two-device LAN**; the 5% loss was **simulated at the sender**, not measured physical network loss. The protocol lacks authentication, encryption, congestion control, adaptive timeouts and crash-resume; finite retry limits apply. Biometric ZIP contents must remain private and must not be committed to GitHub. Instructor approval and any required physical LAN demonstration remain pending.

Evidence: GitHub repository **Anishpantham/BioNet**; Phase 3 reports and CSVs in `docs/` and `results/phase3/`; local terminal sender/receiver logs and independent `cmp` output.